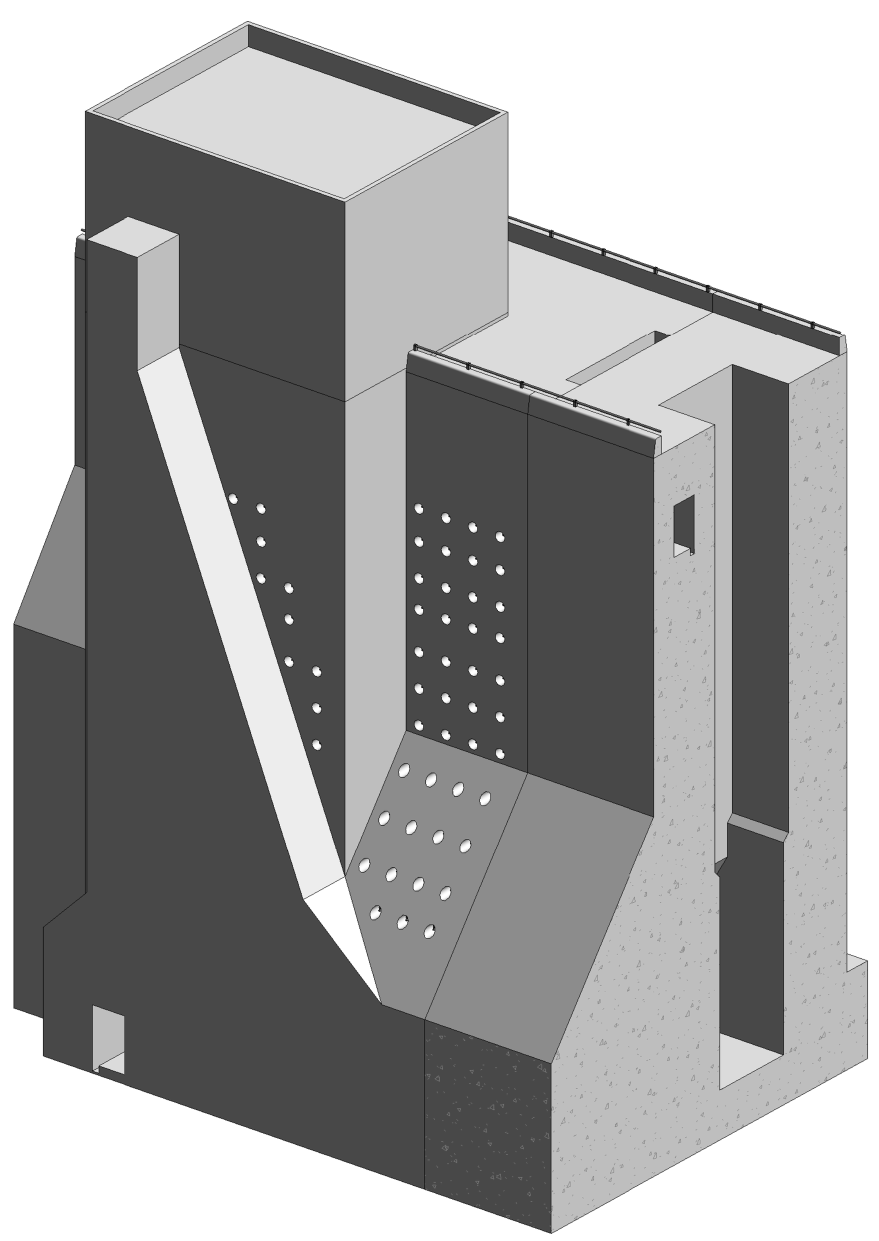
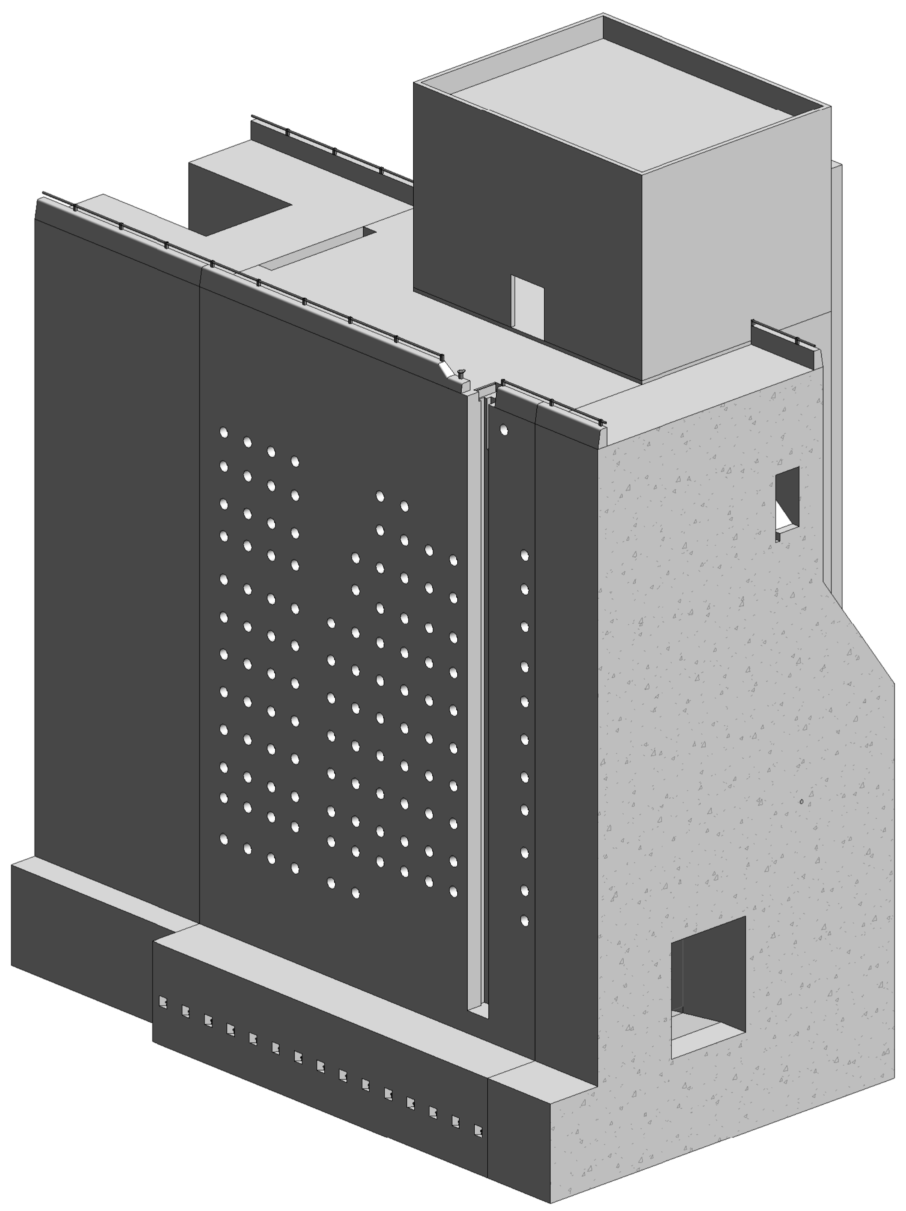
US Army Corps of Engineers ®
MOBILE DISTRICT

HOLT LOCK AND DAM
HOLT LOCK MONOLITH R14 RESTORATION
CHC24010





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GENERAL SCOPE OF WORK - INCLUDES BUT IS NOT LIMITED TO:

BASE CONTRACT

1. PROVIDE A TEMPORARY OFFICE SPACE AND SET UP CONTRACTOR WORK/STAGING AREA AS INDICATED ON CIVIL/SITE DRAWINGS.
2. PROVIDE CONTRACTORS INSTRUMENTATION MONITORING PLAN FOR MONOLITH 14R.
3. IMPLEMENT MONITORING PLAN.
4. LOCALLY DEWATER UPPER RIVER SIDE CULVERT VALVE FOR CULVERT ACCESS.
5. DRILL CORE HOLES AT ANCHOR LOCATIONS AND INSPECT EACH CORE TO DETERMINE DEPTH TO CRACK. INSPECT EACH CORE WITH A DOWN-HOLE CAMERA TO OBSERVE CRACK LOCATION AND CONDITIONS.
6. INSTALL, POST-TENSION, AND PRESSURE GROUT ANCHORS ABOVE LOWER POOL ELEVATION IN MONOLITH 14R AS INDICATED IN DRAWINGS.
7. LOAD TEST ANCHORS
8. EXCAVATE DEBRIS AND DREDGE SILT/SEDIMENTATION FROM LOCK CHAMBER FLOOR IN FRONT OF MONOLITH 14R.
9. INSTALL GRAVEL, FORMWORK, REINFORCING, AND TREMIE CONCRETE PLACEMENTS AS SHOWN ON THE DRAWINGS FOR THE TOE BLOCK.
10. INSTALL LOCK STOPLOGS AND DEWATER LOCK CHAMBER.
11. PUMP OUT ANY REMAINING WATER ON LOCK CHAMBER FLOOR IN WORK AREA.
12. REMOVE TOE BLOCK FORMWORK.
13. INSTALL AND POST-TENSION BASE ANCHORS INTO TOE BLOCK.
14. LOAD TEST TOE BLOCK ANCHORS.
15. INJECT GROUT INTO SURFACE CRACKS AND CULVERT CRACKS.
16. INSTALL ANCHOR CAPS AND GROUT ANCHOR RECESSES.
17. REPAIR METAL WALL ARMORING WHERE DAMAGED DURING CORING.
18. DURING LOCK CHAMBER DEWATERING, CONTRACTOR TO PROVIDE SUPPORT FOR USACE ENGINEERING CULVERT INSPECTIONS.
19. REMOVE LOCK STOPLOGS AND REWATER LOCK CHAMBER.
20. INSTALL WATER STOP AS INDICATED IN THE DRAWINGS.
21. AFTER CONSTRUCTION IS COMPLETE AND THE LOCK CHAMBER IS FILLED TO UPPER POOL, REMOVE AND STORE TEMPORARY CULVERT INTAKE BULKHEAD. BULKHEAD TO BE SET ON A GOVERNMENT BARGE.
22. REMOVE TEMPORARY OFFICE SPACE AND RESTORE SITE TO ORIGINAL CONDITION.

BID OPTION

1. REMOVE RIVERSIDE CULVERT FILLING VALVE FOR CULVERT ACCESS. SET VALVE ON GOVERNMENT BARGE.

SUBMITTED BY:
TECHNICAL LEAD

APPROVAL RECOMMENDED BY:
CHIEF, ENGINEERING DIVISION

APPROVED BY:
DIRECTOR, INDC

THIS PROJECT WAS DESIGNED BY THE MOBILE DISTRICT AND LOUISVILLE DISTRICT OF THE U.S. ARMY CORPS OF ENGINEERS IN COORDINATION WITH THE INLAND NAVIGATION DESIGN CENTER. THE INITIALS OR SIGNATURES AND REGISTRATION DESIGNATIONS OF INDIVIDUALS APPEAR ON THESE PROJECT DOCUMENTS WITHIN THE SCOPE OF THEIR EMPLOYMENT AS REQUIRED BY ER 1110-1-8152.

SOLICITATION NO.:
CONTRACT NO.:
ISSUE DATE: MAY 2025
AMENDMENT 0002 2

US Army Corps of Engineers ®

MARK	DESCRIPTION	DATE
2	AMENDMENT 0002	5/30/25
1	AMENDMENT 0001	5/14/25

US ARMY CORPS OF ENGINEERS
MOBILE DISTRICT
MOBILE, ALABAMA

DESIGNED BY: R. ZUELZKE
DRAWN BY: T. SANDIFER
CHECKED BY: A. JUSTICE
SUBMITTED BY: E. VINCENT
SIZE: ANSI D
ISSUE DATE: MAY 2025
SOLICITATION NO.:
CONTRACT NO.:
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FILE NAME: STRC_Revit Project Template_CESAM 2022.rvt

HOLT LOCK AND DAM
HOLT LOCK MONOLITH R14 RESTORATION

COVER SHEET

SHEET ID
G-001

AMENDMENT 0002

CONTRACTOR ACCESS

1. CONTRACTOR ACCESS VIA LAND WILL BE AT THE MAIN GATE.
2. CONTRACTOR WILL HAVE ACCESS TO A PUBLIC BOAT RAMP APPROXIMATELY HALF A MILE UPSTREAM OF THE WORKSITE.

GENERAL STRUCTURAL NOTES

1. GENERAL NOTES SHALL APPLY UNLESS NOTED OTHERWISE IN THE CONTRACT DOCUMENTS, INCLUDING PLANS, SPECS OR DETAILS. USE TYPICAL DETAILS WHERE APPLICABLE.
2. THE CONTRACTOR MUST REPLACE, AT THEIR OWN EXPENSE, ANY ITEM NOT SPECIFICALLY INDICATED FOR REMOVAL THAT IS DAMAGED OR DESTROYED BY THEIR OPERATIONS.
3. DO NOT SCALE THE DRAWINGS.
4. CONTRACTOR IS RESPONSIBLE TO VERIFY ALL DIMENSIONS AND LOCATE UTILITY CONDUITS AND REINFORCING IN THE FIELD PRIOR TO CONSTRUCTION. REFER TO AS-BUILTS. REPORT DISCREPANCIES TO THE CONTRACTING OFFICER PRIOR TO BEGINNING WORK IN AREAS THAT WILL BE AFFECTED.
5. DETAILS NOT SPECIFICALLY INDICATED SHALL BE SIMILAR TO DETAILS SHOWN FOR SIMILAR CONDITIONS.
6. THE CONTRACTOR SHALL BE SOLELY RESPONSIBLE FOR THE SAFETY OF PERSONS AND PROPERTY DURING CONSTRUCTION.
7. THE CONTRACTOR SHALL BE SOLELY RESPONSIBLE FOR MEANS AND METHODS TO CONSTRUCT THE STRUCTURE, INCLUDING, BUT NOT LIMITED TO, VERIFICATION OF LOAD CAPACITY OF THE STRUCTURE, NEW OR EXISTING, TO SUPPORT CONSTRUCTION ACTIVITIES, EQUIPMENT, ETC. AND LIMITING THE AMOUNT OF CONSTRUCTION LOAD IMPOSED ON THE STRUCTURAL SYSTEM, EXCEPT AS OTHERWISE NOTED IN THE CONTRACT DOCUMENTS. CONSTRUCTION LOADS SHALL NOT EXCEED THE DESIGN CAPACITY OF THE STRUCTURE AT THE TIME THE LOADS ARE IMPOSED. DAMAGE TO THE STRUCTURE CAUSED BY CONSTRUCTION ACTIVITIES SHALL BE CORRECTED BY THE CONTRACTOR AT NO ADDITIONAL COST TO THE GOVERNMENT.
8. REPRODUCTION OF CONTRACT DRAWINGS SHALL NOT BE USED AS SHOP DRAWINGS. SHOP DRAWINGS SHALL BE CHECKED, APPROVED, AND STAMPED BY A PROFESSIONAL ENGINEER PRIOR TO SUBMISSION.
9. USE MOST RECENT VERSION OF REFERENCED AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARD UNLESS SPECIFIED OTHERWISE IN THE REFERENCED UFC, IBC, OR OTHER REFERENCED REQUIREMENT/SPECIFICATIONS.
10. VERIFY ALL DIMENSIONS IN THE FIELD PRIOR TO CONSTRUCTION. REPORT DISCREPANCIES IN DIMENSIONS BETWEEN DIFFERENT DRAWINGS TO THE CONTRACTING OFFICER PRIOR TO BEGINNING WORK IN AREAS THAT WILL BE AFFECTED.
11. UNLESS NOTED OTHERWISE, CALLOUTS FOR TENDON, ANCHOR, AND BAR ALL REFERENCE COLD-JOINT ANCHORS AND THE RELATED HARDWARE DESCRIBED ON SHEET S-501.
12. GOVERNING CODES:
    EM 1110-2-2104 STRENGTH DESIGN FOR REINFORCED CONCRETE HYDRAULIC STRUCTURES
    AISC MANUAL OF STEEL CONSTRUCTION, FIFTEENTH EDITION
    ACI 318-19 "BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE AND COMMENTARY"
    AWS D1.1 STRUCTURAL WELDING CODE
    EM 385-1-1, SAFETY AND HEALTH REQUIREMENTS MANUAL
    PTI TAB.1-06: POST-TENSIONING MANUAL, SIXTH EDITION

DEMOLITION

1. CONTROL DUST RESULTING FROM DEMOLITION TO PREVENT THE SPREAD OF DUST AND AVOID CREATION OF A NUISANCE IN THE SURROUNDING AREA. DO NOT USE WATER WHEN IT WILL RESULT IN, OR CREATE, HAZARDOUS OR OBJECTIONABLE CONDITIONS SUCH AS ICE, FLOODING, OR POLLUTION.
2. UNDAMAGED REINFORCEMENT IS TO BE PROTECTED. REINFORCEMENT EXPOSED BEFORE CONSTRUCTION ACTIVITIES IS TO BE CLEANED BY AIR BLASTING. REINFORCEMENT EXPOSED DURING CONSTRUCTION IS TO BE COATED WITH A RUST INHIBITOR.

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ELEVATION GENERAL NOTES

1. THE CONTRACT DRAWING ELEVATIONS ARE BASED ON THE NATIONAL GEODETIC VERTICAL DATUM, 1929.
2. THE CONTRACTOR SHALL UTILIZE THE USGS RIVER STAGE WEBSITE FOR POOL ELEVATIONS OF HOLT RESERVOIR AND HOLT TAILWATER.
3. THE 50% EXCEEDANCE POOL ELEVATIONS ARE THE MEDIAN POOL ELEVATIONS FROM DATA COLLECTED OVER THE PAST 10 YEARS ACROSS ALL SEASONS. FOR SEASON SPECIFIC DATA, SEE HYDROGRAPHS IN APPENDIX OF SPECIFICATIONS.

CONCRETE GENERAL NOTES

1. ALL REPLACEMENT CONCRETE UNDER THIS CONTRACT SHALL BE CONSIDERED STRUCTURAL CONCRETE.
2. STRUCTURAL CONCRETE DESIGN AND CONSTRUCTION SHALL CONFORM WITH THE AMERICAN CONCRETE INSTITUTE (ACI) 2019 "BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE" (ACI 318), 2020 "SPECIFICATIONS FOR CONCRETE CONSTRUCTION" (ACI 301), AND 2020 "ACI DETAILING MANUAL" - MNL-66(20), EXCEPT AS OTHERWISE REQUIRED BY THE US ARMY CORPS OF ENGINEERS ENGINEERING MANUAL (USACE EM) 1110-2-2104 "STRENGTH DESIGN FOR REINFORCED CONCRETE HYDRAULIC STRUCTURES".
3. CONCRETE SHALL HAVE THE MINIMUM COMPRESSIVE STRENGTH AT 28 DAYS: 4000 PSI WITH A MAXIMUM WATER/CEMENT RATIO = 0.45 AND WITH AN ENTRAINED AIR ADMIXTURE CONFORMING WITH ASTM C260 FOR ALL CONCRETE PERMANENTLY EXPOSED TO THE WEATHER. THE AMOUNT OF ENTRAINED AIR SHALL BE 6% +/- 1%.
4. CALCIUM CHLORIDE SHALL NOT BE PERMITTED NOR SHALL ANY ADMIXTURE CONTAINING CALCIUM CHLORIDE BE PERMITTED.
5. PROPORTION AND DESIGN CONCRETE MIXES TO RESULT IN A CONCRETE SLUMP AT POINT OF PLACEMENT AS INDICATED IN 03 30 53 CAST-IN-PLACE CONCRETE.
6. ALL CONCRETE SHALL CONTAIN A WATER REDUCING ADMIXTURE CONFORMING TO ASTM C494, TYPE A, F OR G.
7. CONCRETE SHALL BE DISCHARGED AT THE SITE WITHIN 1.5 HOURS AFTER WATER HAS BEEN ADDED TO THE CEMENT AND AGGREGATES. ADDITION OF WATER TO THE MIX AT THE PROJECT SITE WILL NOT BE ALLOWED. ALL WATER MUST BE ADDED AT THE BATCH PLANT.
8. STRUCTURAL CONCRETE DENSITY SHALL BE NORMALWEIGHT UNLESS NOTED OTHERWISE.
9. PROVIDE SUPPORTS FOR REINFORCEMENT AND EMBEDMENTS AS NEEDED.
10. PROVIDE 1" CHAMFER AT EXPOSED CONCRETE EDGES OR WHERE CHAMFER IS INDICATED.
11. HOT WEATHER CONCRETE PLACING PER SPEC SECTION 03 30 53
12. SPECIAL WEATHER PROTECTION MEASURES PER SPEC SECTION 03 30 53
13. CONCRETE FIELD QUALITY CONTROL:
    A. TESTING REQUIREMENTS AND REPORTING PER SPEC 03 30 53.
    B. CYLINDER TESTING PER SPEC 03 30 53.

CONCRETE CONSTRUCTION NOTES (SPEC SECTION 03 70 00):

1. ALL CONCRETE CONSTRUCTION SHALL BE IN ACCORDANCE WITH THE SPECIFICATIONS.ALL CONCRETE DESIGN PERFORMED BY THE CONTRACTOR SHALL BE IN ACCORDANCE WITH ACI 318.
2. ALL CAST-IN-PLACE CONCRETE SHALL ATTAIN A MINIMUM 28-DAY COMPRESSIVE STRENGTH (F'C) OF 4,000 PSI.
3. CONCRETE REINFORCING STEEL SHALL BE DEFORMED BARS CONFORMING TO ASTM A615/A615M, GRADE 60 AND SHALL HAVE FABRICATION TOLERANCES IN ACCORDANCE WITH ACI 315.
4. CONCRETE REINFORCING STEEL SHALL BE CONTINUOUS UNLESS OTHERWISE INDICATED. CONTINUOUS REINFORCING STEEL SHALL BE LAPPED IN ACCORDANCE WITH THE REQUIREMENTS OF ACI 318. REINFORCING STEEL SHALL BE PLACED SO THAT SPACING IS GREATER THAN OR EQUAL TO 2X BAR DIAMETER, MINIMUM LAP SPLICE FOR #6 BARS SHALL BE 30"
5. MINIMUM CONCRETE COVER FOR REINFORCING STEEL SHALL BE AS INDICATED.
    a. CONCRETE DEPOSITED AGAINST THE GROUND 6 INCH
    b. CONCRETE EXPOSED TO EARTH OR WEATHER 6 INCH
6. CONCRETE REINFORCING STEEL STANDARD HOOKS, SHALL HAVE A 90 DEGREE HOOK A MINIMUM OF 12 BAR DIAMETERS IN LENGTH, UNLESS OTHERWISE NOTED. STIRRUPS, TIES, AND 180 DEGREE HOOKS SHALL CONFORM TO THE REQUIREMENTS OF ACI 318.
7. ALL EMBEDDED ITEMS SHALL BE PROPERLY PLACED, ACCURATELY POSITIONED AND MAINTAINED SECURELY IN PLACE PRIOR TO AND DURING CONCRETE PLACEMENT.
8. REINFORCING STEEL SHALL BE SPREAD AT SLEEVES, ANCHORS, RECESSES AND OTHER EMBEDDED ITEMS UNLESS OTHERWISE INDICATED. REINFORCEMENT SHALL NOT BE CUT TO FACILITATE PLACEMENT OF EMBEDDED ITEMS, UNLESS INDICATED.
9. NO CONCRETE SHALL BE PLACED UNTIL THE OWNER OR THE OWNER'S DESIGNATED REPRESENTATIVE HAS INSPECTED ALL EMBEDDED WORK, INCLUDING REINFORCEMENT.
10. ALL EXPOSED CONCRETE EDGES SHALL BE CHAMFERED 3/4 INCH UON.
11. ALUMINUM SHALL NOT BE PLACED IN DIRECT CONTACT WITH CONCRETE UNLESS EFFECTIVELY COATED OR COVERED TO PREVENT ALUMINUM-CONCRETE REACTION AND ELECTROLYTIC ACTION BETWEEN ALUMINUM AND STEEL.
12. THE MAXIMUM ALLOWABLE TEMPERATURE DIFFERENTIAL FOR MASS CONCRETE PLACEMENTS BELOW WATER SHALL BE 35°F (19°C) BETWEEN THE CENTER AND THE SURFACE.

MASS CONCRETE PLACEMENT REQUIREMENTS

1. THE CONTRACTOR MUST SUBMIT A DETAILED MASS CONCRETE PLACEMENT PLAN TO USACE CESAM-ENDA FOR REVIEW AND APPROVAL PRIOR TO COMMENCING PLACEMENT ACTIVITIES. THE PLAN IS TO ADDRESS UNDERWATER PLACEMENT AND IN-THE-DRY PLACEMENT.
2. THE SUBMITTED PLAN MUST INCLUDE:
    a. A DESCRIPTION OF THE CONSTRUCTION METHODOLOGY AND SEQUENCE.
    b. DETAILS OF FORMWORK, REINFORCEMENT, AND CONCRETE MIX DESIGN.
    c. A TEMPERATURE CONTROL PLAN TO MITIGATE THERMAL EFFECTS DURING PLACEMENT AND CURING.
    d. (2) SCALED DEMONSTRATION TEST PLAN FOR UNDERWATER CONCRETE PLACEMENT
3. THE TEMPERATURE CONTROL PLAN MUST ADDRESS:
    a. MAXIMUM ALLOWABLE TEMPERATURE DIFFERENTIALS DURING CURING.
    b. PRE-COOLING OR POST-COOLING OF CONCRETE AS NECESSARY.
    c. MONITORING AND LOGGING OF CONCRETE TEMPERATURE USING EMBEDDED SENSORS.
    d. ADJUSTMENTS TO PLACEMENT RATES OR TIMING BASED ON THERMAL DATA.
4. THE CONTRACTOR IS RESPONSIBLE FOR ENSURING ALL PLACEMENT AND CURING PROCESSES MEET USACE SPECIFICATIONS AND PROJECT-SPECIFIC GUIDELINES.
5. ANY MODIFICATIONS TO THE APPROVED PLAN MUST BE SUBMITTED FOR REVIEW AND APPROVAL.
6. ON-SITE INSPECTIONS BY USACE OR DESIGNATED REPRESENTATIVES WILL BE CONDUCTED TO VERIFY COMPLIANCE.
7. THE CONTRACTOR MUST PROVIDE TEMPERATURE MONITORING REPORTS AS PART OF THEIR DAILY LOGS DURING PLACEMENT AND CURING.
8. (2) THE MAXIMUM ALLOWABLE TEMPERATURE DIFFERENTIAL FOR MASS CONCRETE PLACEMENTS BELOW WATER SHALL BE **35°F (19°C)** BETWEEN THE CENTER AND THE SURFACE.
9. PROVIDE SCALED DEMONSTRATION TEST OF UNDERWATER MASS CONCRETE PLACEMENT FOR VERIFICATION OF CONCRETE MIX DESIGN AND PLACEMENT METHOD PRIOR TO PLACEMENT IN THE FIELD.

CONSTRUCTION JOINT PREPARATION

1. CONCRETE SURFACES TO WHICH OTHER CONCRETE IS TO BE BONDED SHALL BE PREPARED FOR RECEIVING THE NEXT LIFT OR ADJACENT CONCRETE BY CLEANING BY SANDBLASTING, HIGH-PRESSURE WATER JET, OR AIR-WATER CUTTING. SURFACE CUTTING BY AIR-WATER JETS WILL NOT BE PERMITTED FOR CONCRETE SURFACES CONGESTED WITH REINFORCING STEEL OR IF THEY ARE RELATIVELY INACCESSIBLE. IF, FOR ANY OTHER REASON, IT IS CONSIDERED UNDESIRABLE TO DISTURB THE SURFACE OF A LIFT BEFORE IT HAS HARDENED, THE USE OF SANDBLASTING OR HIGH-PRESSURE WATER JET AFTER HARDENING WILL BE REQUIRED. REGARDLESS OF THE METHOD USED, THE RESULTING SURFACE SHALL BE FREE FROM ALL LAITANCE AND INFERIOR CONCRETE SO THAT CLEAN, WELL-BONDED COARSE AGGREGATE PARTICLES ARE EXPOSED UNIFORMLY OVER THE LIFT SURFACE.
2. TREATMENT METHOD SHALL BE SUCH THAT THE EDGES OF THE LARGER PARTICLES OF AGGREGATE ARE NOT UNDERCUT. WHERE JOINT PREPARATION OCCURS MORE THAN 2 DAYS PRIOR TO PLACING THE NEXT LIFT OR WHERE THE WORK IN THE AREA SUBSEQUENT TO THE JOINT PREPARATION CAUSES DIRT OR DEBRIS TO BE DEPOSITED ON THE SURFACE, THE SURFACE SHALL BE CLEANED AS THE LAST OPERATION PRIOR TO PLACING THE NEXT LIFT. THE SURFACE OF THE CONSTRUCTION JOINT SHALL BE KEPT CONTINUOUSLY WET FOR THE FIRST 12 HOURS OF THE 24 HOURS PRIOR TO PLACING CONCRETE, EXCEPT THAT THE SURFACE SHALL BE DAMP WITH NO FREE WATER AT THE TIME OF PLACEMENT.
3. HIGH-PRESSURE WATER JET: A STREAM OF WATER UNDER A PRESSURE OF NOT LESS THAN 3,000 PSI MAY BE USED FOR CLEANING. ITS USE SHALL BE DELAYED UNTIL THE CONCRETE IS SUFFICIENTLY HARD SO THAT ONLY THE SURFACE SKIN OR MORTAR IS REMOVED AND THERE IS NO UNDERCUTTING OF COARSE-AGGREGATE PARTICLES. IF THE HIGH-PRESSURE WATER JET IS INCAPABLE OF A SATISFACTORY CLEANING, THE SURFACE SHALL BE CLEANED BY SANDBLASTING.
4. WET SANDBLASTING: THIS METHOD OF JOINT PREPARATION MAY BE USED WHEN THE CONCRETE HAS REACHED SUFFICIENT STRENGTH TO PREVENT UNDERCUTTING OF COARSE AGGREGATE PARTICLES.
5. THE OPERATION SHALL BE CONTINUED UNTIL ALL ACCUMULATED LAITANCE, COATINGS, STAINS, DEBRIS, AND FOREIGN MATERIALS ARE REMOVED. THE SURFACE OF THE CONCRETE SHALL THEN BE WASHED THOROUGHLY TO REMOVE ALL LOOSE MATERIAL.
6. THIS METHOD MAY BE USED ON BOTH HORIZONTAL AND VERTICAL SURFACES.

UNWATERING

THE LOCK CHAMBER WILL REQUIRE UNWATERING TO COMPLETE THIS CONTRACT. CONTRACTOR MUST COORDINATE WITH LOCK SUPERVISOR AND COR TO UNWATER LOCK CHAMBER.

CHAMBER ACCESS RESTRICTIONS

SEE SPECIFICATION SECTION 01 00 00 FOR CHAMBER CLOSURE RESTRICTIONS.

SITE UTILITY ACCESS

1. RAW WATER FOR CONSTRUCTION PURPOSES MAY BE FILLED ON-SITE FROM THE FIRE PUMP SYSTEM. COORDINATE SYSTEM REQUIREMENTS AND LOGISTICS WITH THE LOCK SUPERVISOR.
2. CONTRACTORS MAY TAP INTO ON-SITE ELECTRICAL OUTLETS, PROVIDED THEY ARE AVAILABLE, DOING SO DOES NOT CREATE A TRIPPING HAZARD TO PERSONNEL, AND THEY ARE USED WITHIN THEIR OPERATIONAL LIMITS.

GROUT AND EPOXY GENERAL NOTES

1. SURFACE CRACKS IN ALL MONOLITHS IN THE CONTRACT SHALL BE SEALED WITH INJECTED EPOXY OR POLYURETHANE GROUT PER SPEC SECTION 03 01 32.
2. VERTICAL CRACK IN MONOLITH 14R SHALL BE FILLED VIA PRESSURE GROUTING WITH CEMENT GROUT CONTAINING A CRYSTALLINE ADMIXTURE PER SPEC SECTION 03 01 30.72 OR 31 68 13.

MUD AND DEBRIS REMOVAL

MUD AND DEBRIS MAY BE RELOCATED WITHIN THE CHAMBER AS APPROVED BY THE COR. CONTRACTOR SHALL NOT STACK RELOCATED MUD AND DEBRIS HIGHER THAN THE TOP OF TOE BLOCK AT ANY LOCATION WITHIN THE LOCK CHAMBER.

SUBMITTALS

1. REFER TO SPECIFICATIONS FOR FULL LIST OF PROJECT SUBMITTALS REQUIRED.
2. SUBMIT THE METHOD AND EQUIPMENT PROPOSED FOR JOINT CLEANUP AND WASTE DISPOSAL, FOR REVIEW 30 DAYS BEFORE CONCRETE PLACEMENT BEGINS.
3. SUBMIT CONCRETE MIX DESIGN WITH COMPLETE STATISTICAL BACKUP.
4. SUBMIT THE CONCRETE TEST REPORTS.
5. SUBMIT A FLOATING PLANT LAYOUT PLAN.
6. SUBMIT MANUFACTURER PRODUCT DATA SHEETS FOR ROCK ANCHORS AND WATERSTOPS.
7. SUBMIT CHEMICAL GROUTING PLAN.
8. ANCHOR INSTALLATION WORK PLAN

POST-INSTALLED DOWEL NOTES

1. PROVIDE ANCHORS OF THE TYPE, EFFECTIVE EMBEDMENT, DIAMETER, AND SPACING AS INDICATED. ADDITIONAL REQUIREMENTS ARE DEFINED IN THE FOLLOWING NOTES.
2. STEEL ANCHOR ELEMENTS: REBAR DOWEL ANCHORS MUST MEET THE REQUIREMENTS OF ASTM A615 GRADE 60.
3. ADHESIVE: UNLESS SPECIFICALLY INDICATED OTHERWISE, ALL POST-INSTALLED ANCHORING ELEMENTS ARE TO BE INSTALLED WITH AN ADHESIVE ANCHORING SYSTEM. ADHESIVE MUST BE A STRUCTURAL ADHESIVE THAT IS RATED FOR SUBMERGED/UNDERWATER INSTALLATIONS. ANCHORS MUST HAVE BEEN TESTED AND QUALIFIED FOR PERFORMANCE IN CRACKED AND UNCRACKED CONCRETE, HORIZONTAL AND OVERHEAD APPLICATIONS, AND LONG TERM CREEP IN ACCORDANCE WITH ACI 355.4.
4. ADHESIVE ANCHORS MUST HAVE THE BELOW CHARACTERISTIC BOND STRENGTHS FOR UNCRACKED AND CRACKED CONCRETE, ASSUMING 3,000 PSI CONCRETE AND ASSUMING MAXIMUM SHORT TERM TEMPERATURES OF 130°F AND MAXIMUM LONG TERM TEMPERATURES OF 110°F.

REBAR DIAMETER (IN)	Tau, uncr (PSI)	Tau, cr (PSI)
3/8	1,530	820
1/2	1,500	830
5/8	1,470	830
3/4	1,430	840
7/8	1,400	850
1	1,370	860

Tau, uncr = CHARACTERISTIC BOND STRENGTH, UNCRACKED CONCRETE.
Tau, cr = CHARACTERISTIC BOND STRENGTH, CRACKED CONCRETE.

5. SUBMIT MANUFACTURERS DATA AND ICC-ES EVALUATION REPORT, OR THIRD PARTY EVALUATION REPORT IN CONFORMANCE WITH ACI 355.2 OR ACI 355.4 FOR APPROVAL. THIRD PARTY MUST BE ACCREDITED UNDER ISO/IEC 17025 BY A RECOGNIZED ACCREDITATION BODY CONFORMING TO THE REQUIREMENTS OF ISO/IEC 17011 IN ACCORDANCE WITH ACI 355.4. SUBMITTAL MUST INCLUDE MANUFACTURER'S INSTALLATION INSTRUCTIONS.

ABBREVIATIONS

APPROX = APPROXIMATELY
B/ = BOTTOM
CL = CENTERLINE
CONC = CONCRETE
COR = CONTRACTING OFFICER'S REPRESENTATIVE
CORR = CORROSIVE
CLR = CLEAR
HORZ = HORIZONTAL
ELEV = ELEVATION
EXIST = EXISTING
HP =HIGH POINT
ICC =INTERNATIONAL CODE COUNCIL
LLH = LONG LEG HORIZONTAL
OAL = OVERALL LENGTH
OC = ON CENTER
OPNG = OPENING
PL = PLATE
PLCS = PLACES
SIM =SIMILAR
SPA = SPACING
T/ = TOP
TYP = TYPICAL
VERT = VERTICAL
W/ = WITH

GENERAL HATCH PATTERN LEGEND

SECTION TAKEN THROUGH CONCRETE MATERIAL
SECTION TAKEN THROUGH GROUT MATERIAL
SECTION TAKEN THROUGH STEEL MATERIAL
SECTION TAKEN THROUGH REINFORCEMENT
CONCRETE DEMOLITION AREA
CONCRETE REPAIR AREA

US Army Corps of Engineers ®

MARK	DESCRIPTION	DATE
2	AMENDMENT 0002	5/30/25
1	AMENDMENT 0001	5/14/25

US ARMY CORPS OF ENGINEERS
MOBILE DISTRICT
MOBILE, ALABAMA

DESIGNED BY: R. ZUELZKE
DRAWN BY: T. SANDIFER
CHECKED BY: A. JUSTICE
SUBMITTED BY: E. VINCENT
SIZE: ANSI D
FILE NAME: STRC_Revit Project Template_CESAM 2022.rvt
ISSUE DATE: MAY 2025
SOLICITATION NO.:
CONTRACT NO.:
PROJECT NO.: CHC24010

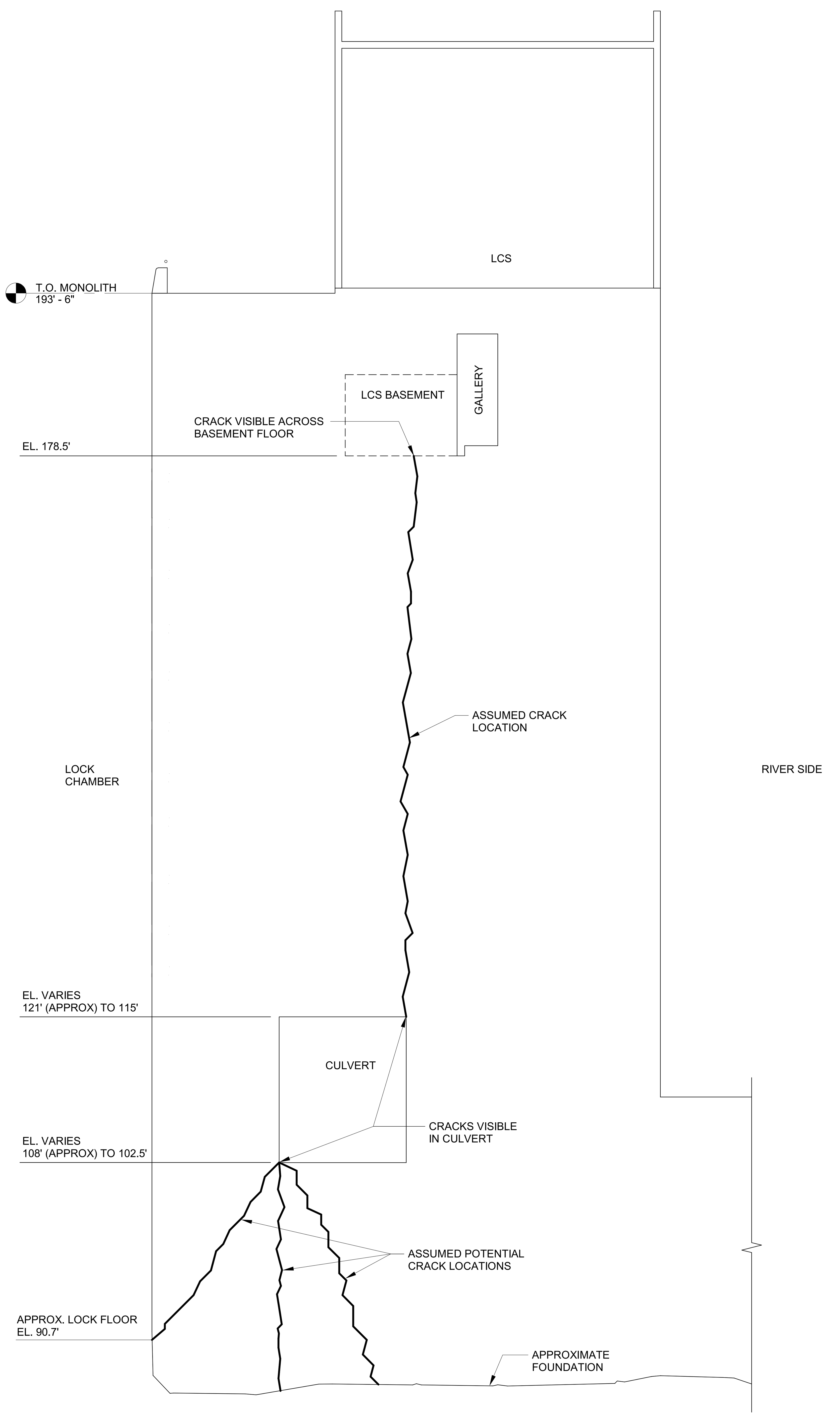
HOLT LOCK AND DAM
HOLT LOCK MONOLITH R14 RESTORATION

GENERAL NOTES, ABBREVIATIONS

SHEET ID

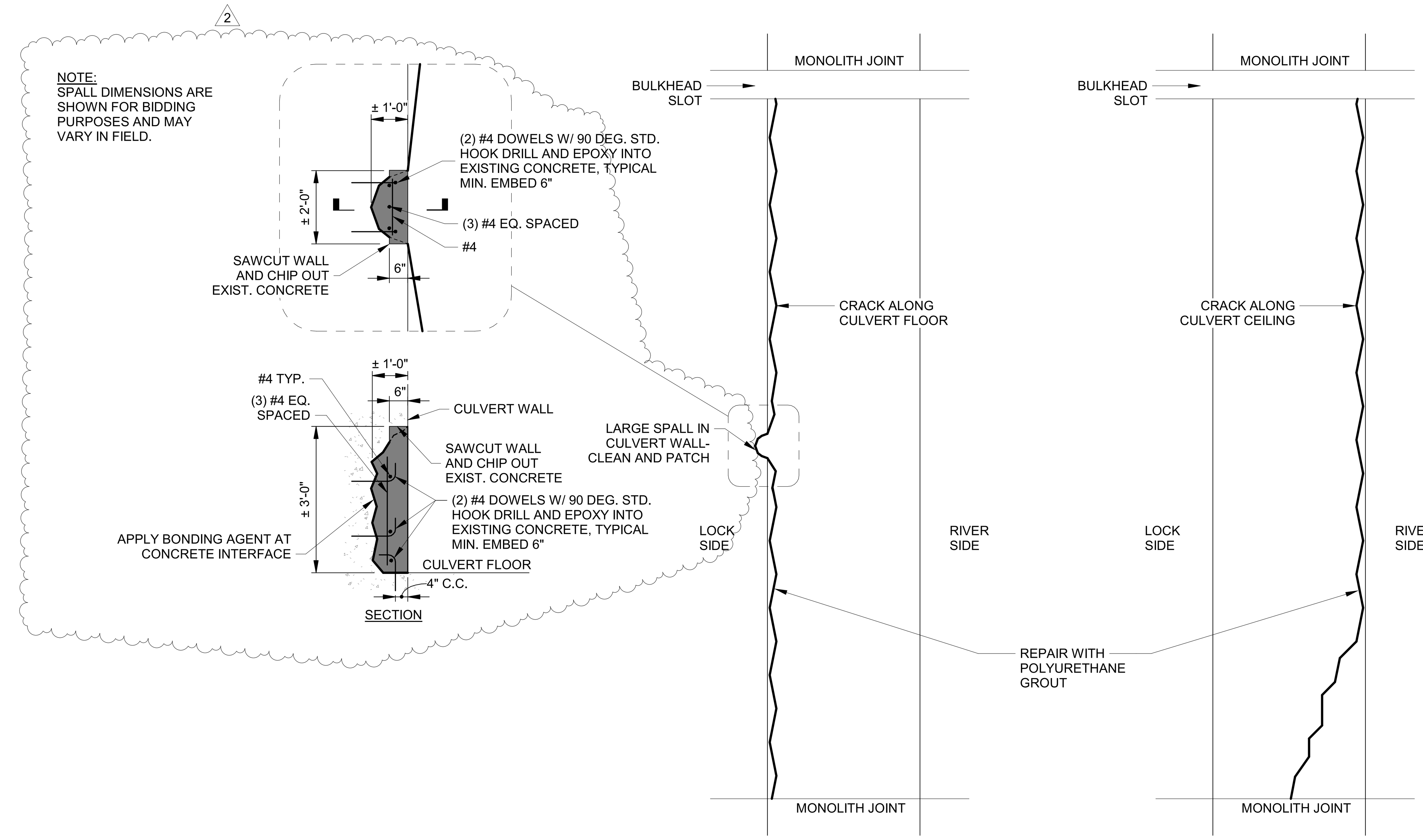
G-002

AMENDMENT 0002



1 14R UPSTREAM ELEVATION

1/8" = 1'-0"



2 CULVERT FLOOR AND CEILING SURFACE CRACKS

1/8" = 1'-0"

US Army Corps of Engineers ®

MARK	DESCRIPTION	DATE
2	AMENDMENT 0002	5/30/25
1	AMENDMENT 0001	5/14/25

DESIGNED BY: R. ZUELZKE
DRAWN BY: T. SANDIFER
CHECKED BY: A. JUSTICE
SUBMITTED BY: E. VINCENT
SIZE: ANSI D
FILE NAME:

ISSUE DATE: MAY 2025
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CONTRACT NO.:
PROJECT NO.: CHC24010

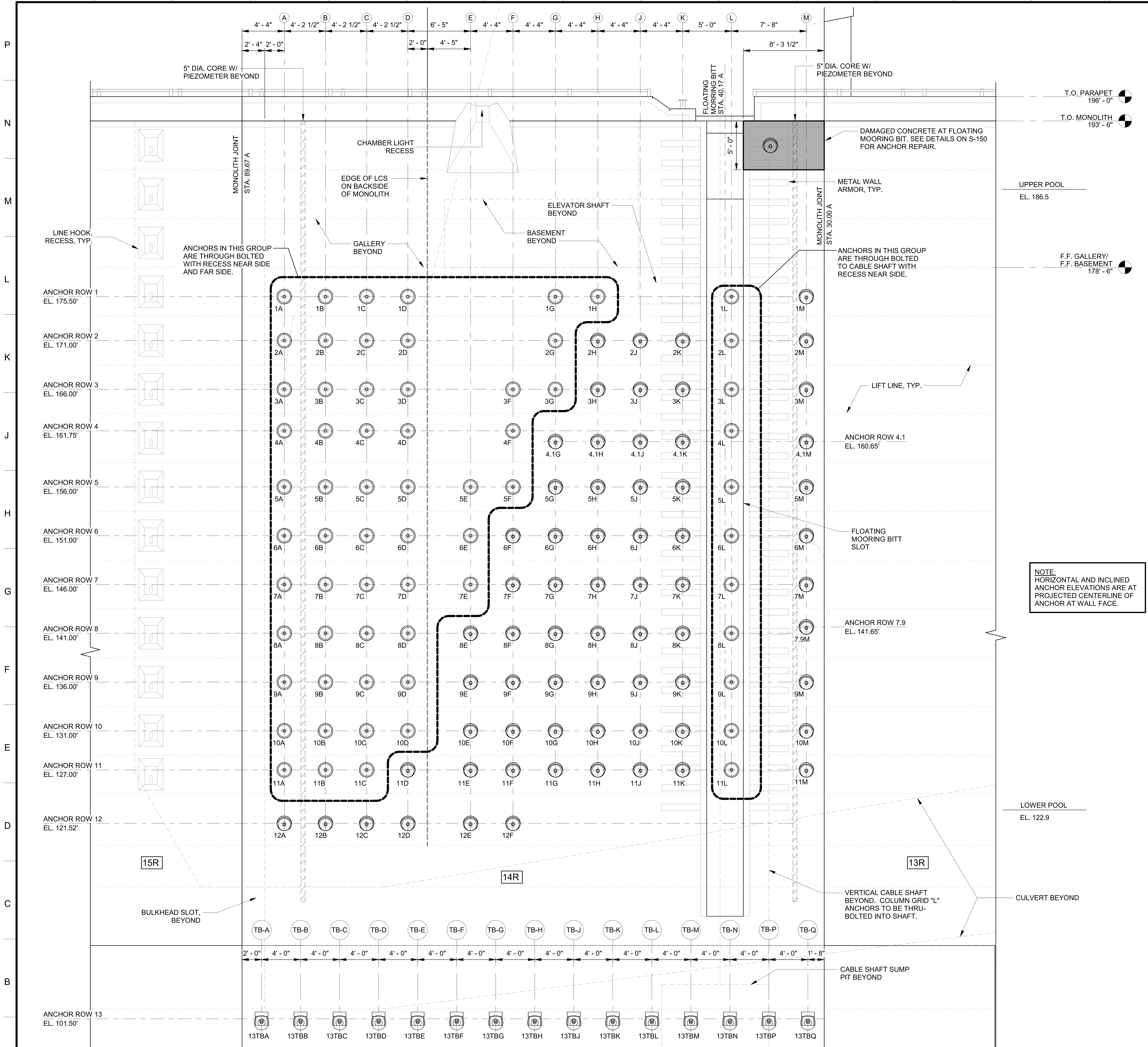
US ARMY CORPS OF ENGINEERS
MOBILE DISTRICT
MOBILE, ALABAMA

HOLT LOCK AND DAM
HOLT LOCK MONOLITH R14 RESTORATION

14R CULVERT CRACK LOCATIONS

SHEET ID
S-113

AMENDMENT 0002



MONOLITH 14R CHAMBER SIDE ELEVATION-ANCHOR LOCATIONS

3/16" = 1'-0"

ANCHOR TABLE

ANCHOR NUMBER	* EFFECTIVE ANCHOR LENGTH	** STRESSING SEQUENCE	HOLE ALIGNMENT TOLERANCE
1A	34'-8"	8	1:200
1B	34'-8"	6	1:57
1C	34'-8"	4	1:57
1D	34'-8"	2	1:57
1G	46'-8"	1	1:57
1H	46'-8"	3	1:200
1L	30'-1"	5	1:200
1M	44'-0"	7	1:200
2A	34'-8"	18	1:200
2B	34'-8"	16	1:57
2C	34'-8"	14	1:57
2D	34'-8"	12	1:57
2G	46'-8"	10	1:57
2H	44'-0"	9	1:57
2J	44'-0"	11	1:57
2K	44'-0"	13	1:57
2L	30'-1"	15	1:200
2M	44'-0"	17	1:200
3A	34'-8"	28	1:200
3B	34'-8"	26	1:57
3C	34'-8"	24	1:57
3D	34'-8"	22	1:57
3F	46'-8"	20	1:57
3G	46'-8"	19	1:100
3H	44'-0"	21	1:57
3J	44'-0"	23	1:57
3K	44'-0"	25	1:57
3L	30'-1"	27	1:200
3M	44'-0"	29	1:200
4A	34'-8"	39	1:200
4B	34'-8"	37	1:57
4C	34'-8"	35	1:57
4D	34'-8"	33	1:57
4F	46'-8"	31	1:200
4.1G	44'-0"	30	1:200
4.1H	44'-0"	32	1:100
4.1J	44'-0"	34	1:100
4.1K	44'-0"	36	1:100
4L	30'-1"	38	1:200
4.1M	44'-0"	40	1:200
5A	34'-8"	52	1:200
5B	34'-8"	50	1:57
5C	34'-8"	48	1:57
5D	34'-8"	46	1:57
5E	46'-8"	44	1:57
5F	46'-8"	42	1:200
5G	44'-0"	41	1:57
5H	44'-0"	43	1:57
5J	44'-0"	45	1:57
5K	44'-0"	47	1:57
5L	30'-1"	49	1:200
5M	44'-0"	51	1:200
6A	34'-8"	64	1:200
6B	34'-8"	62	1:57
6C	34'-8"	60	1:57
6D	34'-8"	58	1:57
6E	46'-8"	56	1:57
6F	44'-0"	54	1:57
6G	44'-0"	53	1:57
6H	44'-0"	55	1:57
6J	44'-0"	57	1:57
6K	44'-0"	59	1:57
6L	30'-1"	61	1:200
6M	44'-0"	63	1:200
7A	34'-8"	76	1:200
7B	34'-8"	74	1:57
7C	34'-8"	72	1:57
7D	34'-8"	70	1:57
7E	46'-8"	68	1:57
7F	44'-0"	66	1:57
7G	44'-0"	65	1:57
7H	44'-0"	67	1:57
7J	44'-0"	69	1:57
7K	44'-0"	71	1:57
7L	30'-1"	73	1:200
7M	44'-0"	75	1:200
8A	37'-0"	88	1:200
8B	37'-0"	86	1:57
8C	37'-0"	84	1:57
8D	37'-0"	82	1:57
8E	44'-0"	80	1:57
8F	44'-0"	78	1:57
8G	44'-0"	77	1:57
8H	44'-0"	79	1:57
8J	44'-0"	81	1:57
8K	44'-0"	83	1:57
8L	30'-1"	85	1:200
7.9M	44'-0"	87	1:200
9A	41'-0"	100	1:200
9B	41'-0"	98	1:57
9C	41'-0"	96	1:57
9D	41'-0"	94	1:57
9E	44'-0"	92	1:57
9F	44'-0"	90	1:57
9G	44'-0"	89	1:57
9H	44'-0"	91	1:57
9J	44'-0"	93	1:57
9K	44'-0"	95	1:57
9L	30'-1"	97	1:200
9M	44'-0"	99	1:200
10A	44'-11"	112	1:200
10B	44'-11"	110	1:57
10C	44'-11"	108	1:57
10D	44'-11"	106	1:57
10E	44'-0"	104	1:57
10F	44'-0"	102	1:57
10G	44'-0"	101	1:57
10H	44'-0"	103	1:57
10J	44'-0"	105	1:57
10K	44'-0"	107	1:57
10L	30'-1"	109	1:200
10M	44'-0"	111	1:200
11A	48'-5"	124	1:200
11B	48'-5"	122	1:57
11C	48'-5"	120	1:200
11D	44'-0"	118	1:200
11E	44'-0"	116	1:200
11F	44'-0"	114	1:200
11G	44'-0"	113	1:200
11H	44'-0"	115	1:200
11J	44'-0"	117	1:200
11K	44'-0"	119	1:150
11L	30'-1"	121	1:200
11M	44'-0"	123	1:200
12A	44'-0"	130	1:200
12B	44'-0"	128	1:57
12C	44'-0"	126	1:57
12D	44'-0"	125	1:57
12E	44'-0"	127	1:57
12F	44'-0"	129	1:200

ANCHOR TABLE

ANCHOR NUMBER	* EFFECTIVE ANCHOR LENGTH	** STRESSING SEQUENCE	HOLE ALIGNMENT TOLERANCE
13TBA	48'-0"	144	1:100
13TBB	48'-0"	142	1:57
13TBC	48'-0"	140	1:57
13TBD	48'-0"	138	1:57
13TBE	48'-0"	136	1:57
13TBF	48'-0"	134	1:57
13TBG	48'-0"	132	1:57
13TBH	48'-0"	131	1:57
13TBJ	48'-0"	133	1:57
13TBK	48'-0"	135	1:57
13TBL	48'-0"	137	1:100
13TBM	48'-0"	139	1:100
13TBN	48'-0"	141	1:100
13TBP	48'-0"	143	1:100
13TBQ	48'-0"	145	1:100
FMB	24'-0"	-	1:100

2

* EFFECTIVE ANCHOR LENGTH IS MEASURED FROM BOTTOM OF BEARING PLATE(S) AND IS APPROXIMATE AND FOR REFERENCE ONLY. CONTRACTOR MUST DESIGN ANCHORAGE AND STRESSING LENGTH. FINAL OVERALL ANCHOR LENGTH PENDING SPECIFIC HARDWARE REQUIREMENTS BY CONTRACTOR.

** ALTERNATE CONTRACTOR-SUPPLIED ANCHOR STRESSING SEQUENCE MAY BE ALLOWED PENDING GOVERNMENT APPROVAL. CONTRACTOR SHALL INSTALL, AT MINIMUM, THRU-BOLTED ANCHORS ON ROWS 1 AND 2 FIRST.

ANCHOR ROD NOTE:

ALL ANCHORS ON ROWS 1-12 SHALL BE 2 1/2"Ø GRADE 150 KSI ALL-THREAD-BAR. ALL ANCHORS ON ROW 13 (TOE BLOCK ANCHORS) SHALL BE 3"Ø GRADE 150 KSI ALL-THREAD-BAR.

US Army Corps of Engineers ®

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MOBILE DISTRICT
MOBILE, ALABAMA

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HOLT LOCK AND DAM
HOLT LOCK MONOLITH R14 RESTORATION

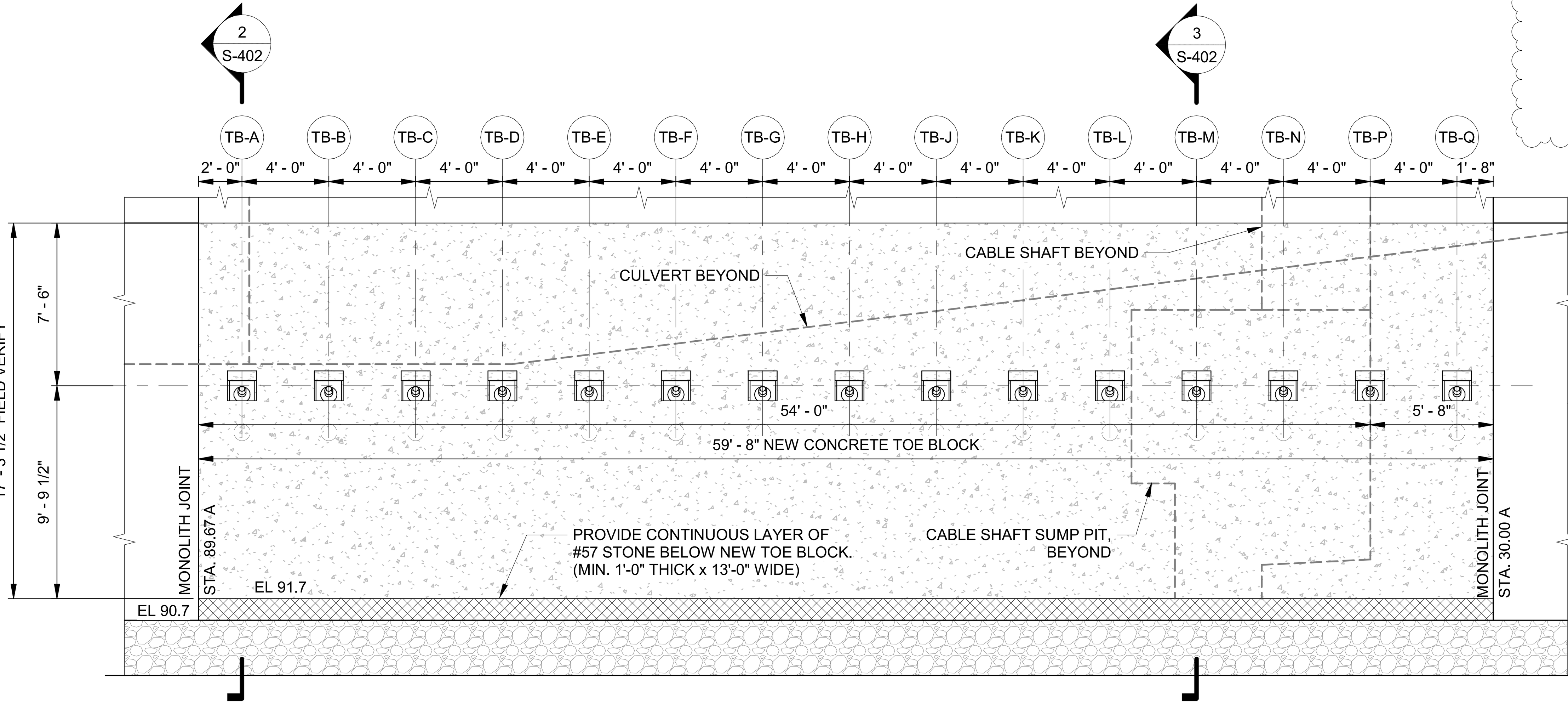
ANCHOR LOCATION ELEVATION

SHEET ID

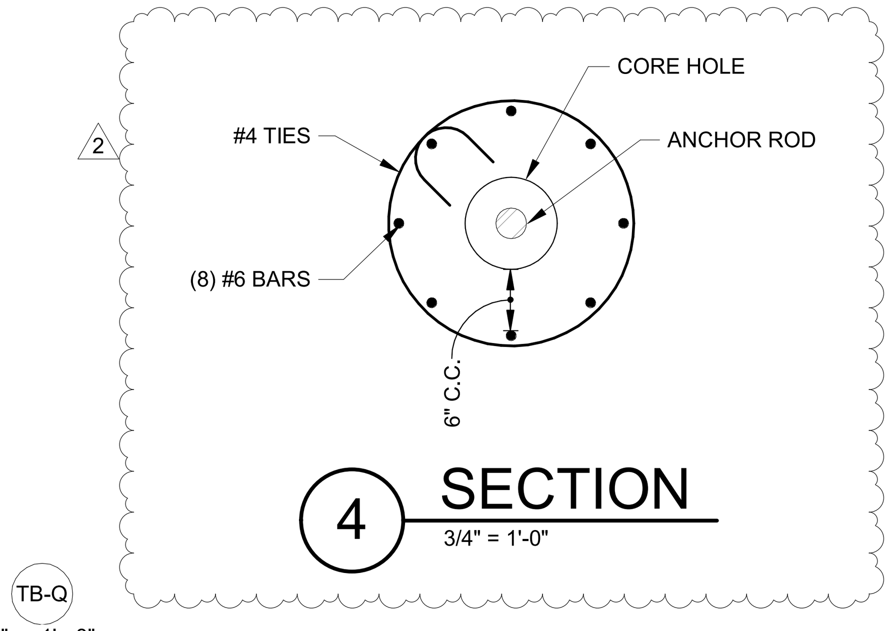
S-131

AMENDMENT 0002

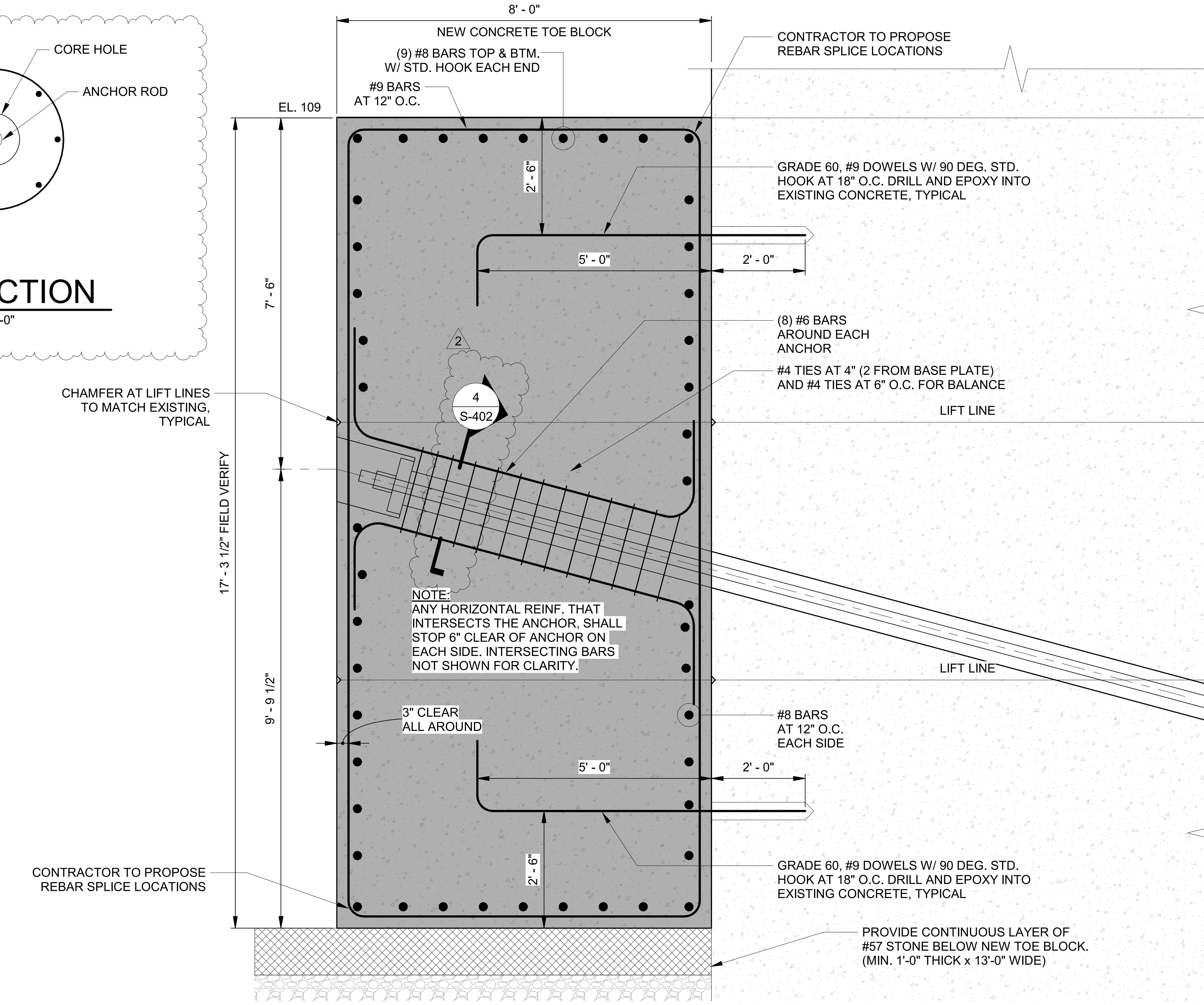
NOTE:
TOE BLOCK SHALL BE CONSTRUCTED IN-THE-WET.



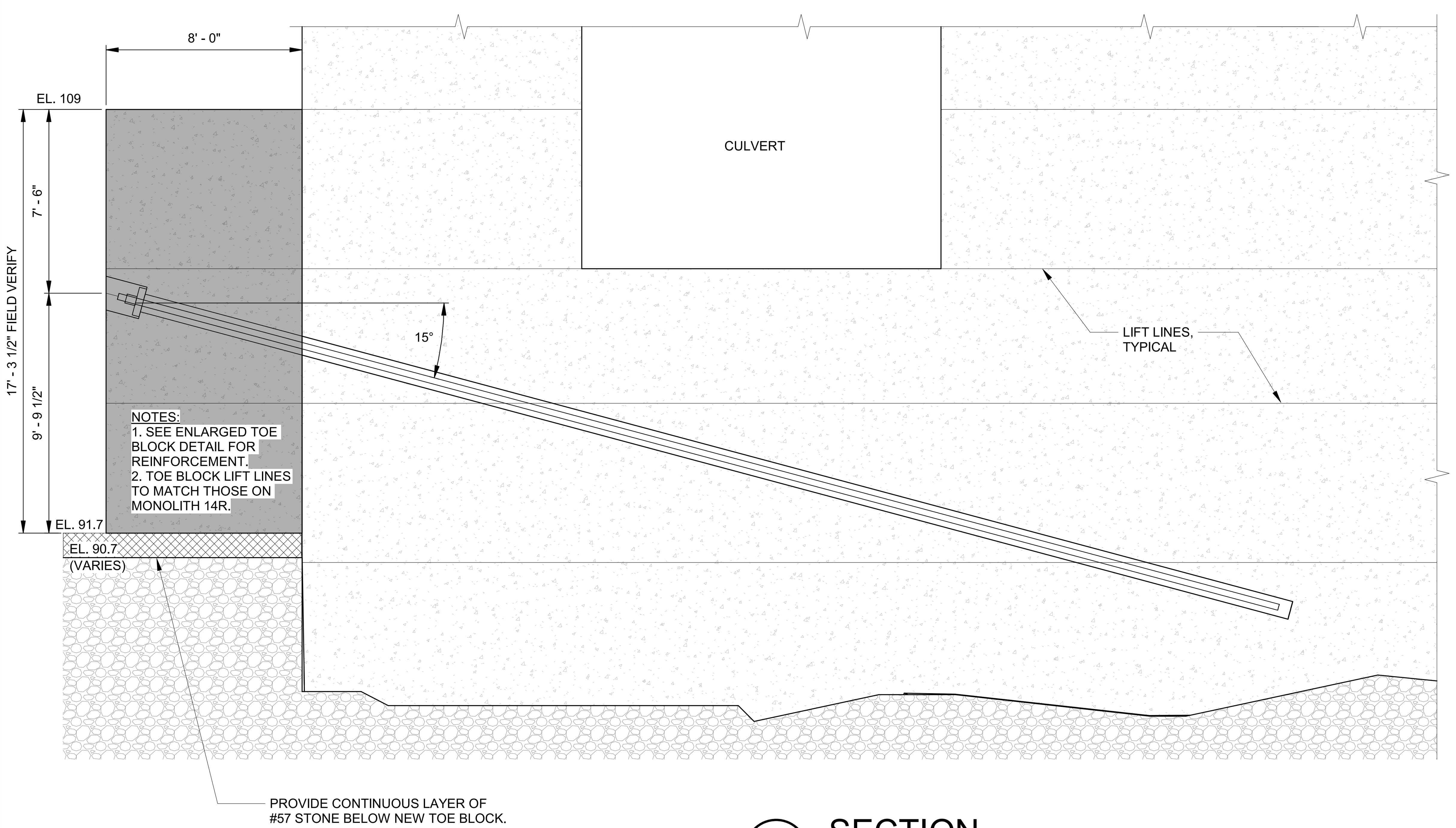
1 MONOLITH 14R CHAMBER SIDE ELEVATION-BASE ANCHOR LOCATIONS
3/16" = 1'-0"



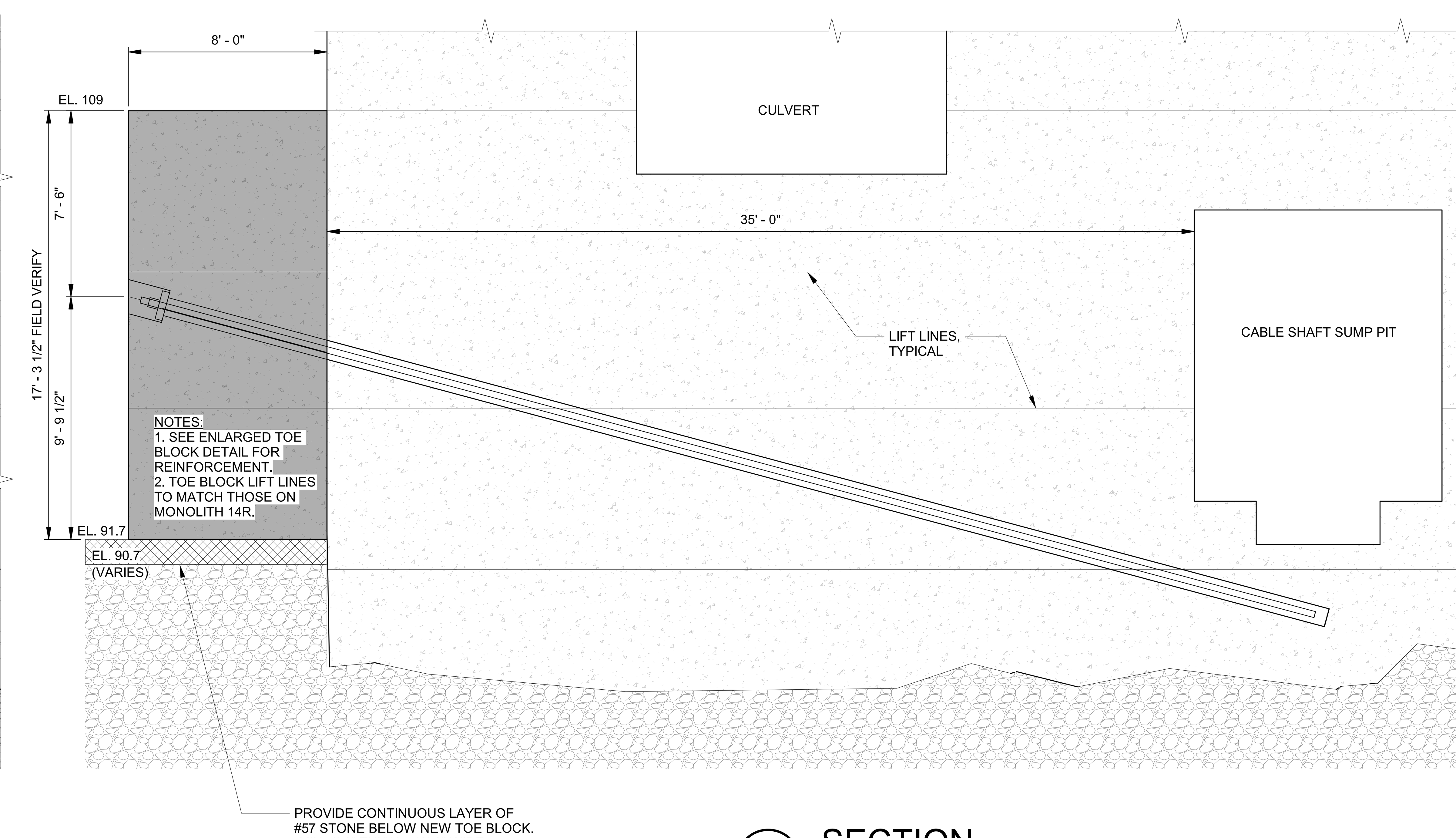
4 SECTION
3/4" = 1'-0"



ENLARGED TOE BLOCK DETAIL
1/2" = 1'-0"



2 SECTION
1/4" = 1'-0"



3 SECTION
1/4" = 1'-0"

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PROJECT NO.: CHC24010
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US ARMY CORPS OF ENGINEERS
MOBILE DISTRICT
MOBILE, ALABAMA

HOLT LOCK AND DAM
HOLT LOCK MONOLITH R14 RESTORATION

TOE BLOCK DETAILS

SHEET ID
S-402

AMENDMENT 0002